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Directing light into an optical fiber

Abstract

A system can direct light into an optical fiber. Imaging optics can form an image of an end of an optical fiber. An actuatable optical element can be configured to define an optical path that extends to the actuatable optical element and further extends to the end of the optical fiber. A processor can determine a location in the image of a specified feature in the image. The processor can cause, based on the location of the specified feature in the image, the actuatable optical element to actuate to align the optical path to a core of the optical fiber. A light source can direct a light beam along the optical path to couple into the core of the optical fiber.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a U.S. National Stage Filing under 35 U.S.C. 371 from International Application No. PCT/US2021/035519, filed on Jun. 2, 2021, and published as WO 2021/247754 A1 on Dec. 9, 2021, which claims the benefit of U.S. Provisional Application No. 63/034,277, filed Jun. 3, 2020, each of which is incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE
(1) The present disclosure relates to an optical system that can direct light into an optical fiber.
BACKGROUND OF THE DISCLOSURE
(2) An optical system can use an optical fiber, such as a single-mode fiber. Misalignment of a light beam, with respect to a core of the single-mode fiber, can reduce the coupling efficiency of the beam into the core, and can increase losses in the optical system.
SUMMARY OF THE INVENTION
(3) In an example, a system can direct light into an optical fiber. The system comprises imaging

optics, an actuatable optical element, a processor, and a light source. The imaging optics is configured to form an image of an end of an optical fiber. The actuatable optical element is configured to define an optical path that extends to the actuatable optical element and further extends to the end of the optical fiber. The processor is configured to determine a location in the image of a specified feature in the image. The processor is further configured to cause, based on the location of the specified feature in the image, the actuatable optical element to actuate to align the optical path to a core of the optical fiber. The light source is configured to direct a light beam along the optical path to couple into the core of the optical fiber.
(4) In another example, a method is for operating a system to direct light into an optical fiber. The system comprises imaging optics, a processor, and an actuatable optical element. The actuatable optical element defines an optical path, the optical path extending to the actuatable optical element and further extending to the end of the optical fiber. The method comprises: generating, with the imaging optics, an image of an end of the optical fiber; determining, with the processor, a location in the image of a specified feature in the image; causing, with the processor, the actuatable optical element to actuate to align the optical path to a core of the optical fiber based on the location of the specified feature in the image; and directing a light beam along the optical path to couple into the core of the optical fiber.
(5) In another example, a computer-readable medium stores instructions that, when executed by a processor of a system for directing light into an optical fiber, can cause the processor to execute operations, such as the method described above or described elsewhere in this description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows a schematic drawing of an example of an apparatus that includes a system for directing light into an optical fiber.
(2) FIG. **2** shows an end-on view of an example of a light source configured as a fiber bundle, which is suitable for use in the system of FIG. **1**.
(3) FIG. **3** shows a top view of an example of a portion of the system of FIG. **1**, in which the objective element is configured as an objective mirror.
(4) FIG. **4** shows a top view of an example of longitudinal position sensor elements that are suitable for use in the system of FIG. **1**.
(5) FIG. **5** shows a top view of another example of longitudinal position sensor elements that are suitable for use in the system of FIG. **1**.
(6) FIG. **6** shows a top view of another example of longitudinal position sensor elements that are suitable for use in the system of FIG. **1**.
(7) FIG. **7** shows a top view of the longitudinal position adjustor elements in the system of FIG. **1**.
(8) FIG. **8** shows a top view of an example of longitudinal position adjustor elements suitable for use in the system of FIG. **1**.
(9) FIG. **9** shows a flowchart of an example of a method for operating a system to direct light into an optical fiber.
(10) FIG. **10** shows a flowchart of an example of another method for operating a system to direct light into an optical fiber.
(11) FIG. **11** is a simplified diagram showing selective components of the system of FIG. **1** to illustrate the use of parabolic mirrors in place of lenses.
(12) FIG. **12** shows a schematic diagram of an example system that allows imaging of both the capital-side optical fiber and the multi-core sensor.
(13) FIG. **13** shows an example image generated by the system of FIG. **12**.
(14) FIG. **14** shows a schematic diagram of an example system for making a non-contact

connection between a capital-side fiber and a multi-core sensor that uses two tip/tilt mirrors for beam steering and alignment.

(15) FIGS. **15** and **16** show a simple model of a system with one and two tip/tilt mirrors, respectively.

(16) Corresponding reference characters indicate corresponding parts throughout the several views. Elements in the drawings are not necessarily drawn to scale. The configurations shown in the drawings are merely examples and should not be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION

(17) in an example, a system can direct light into an optical fiber. Imaging optics can form an image of an end of an optical fiber. An actuatable optical element can define an optical path that extends to the actuatable optical element and further extends to the end of the optical fiber. A processor can determine a location in the image of a specified feature in the image. The processor can cause, based on the location of the specified feature in the image, the actuatable optical element to actuate to align the optical path to a core of the optical fiber. A light source can direct a light beam along the optical path to couple into the core of the optical fiber.

(18) The system can identify a feature in an image of the end of the optical fiber, then use the location of the feature to actively align an optical path to a core of the optical fiber. With active alignment, the system can improve the robustness of the alignment of the light beam to the core of the optical fiber. Improved robustness of alignment can help compensate for misalignments due to physical misalignment of the optical fiber, such as due to manufacturing tolerances, non-idealities in the mounting of a mechanical holder, and so forth. As a result, the system can achieve a higher coupling efficiency of the beam into the core of the optical fiber compared to an otherwise identical system not utilizing this technique.

(19) Further, in various examples, the active alignment can be performed one or more times for each use of a system. Active alignment can be performed before and/or during use of the system. As a specific example, active alignment performed one or more times during operation, can help improve or maintain the alignment while an optical system is in operation. For example, during operation, an optical system can experience movement, temperature change, physical shock or vibration (such as caused by air currents), and/or other environmental or physical change that affects the alignment. The system described below can compensate periodically, in response to a determination of misalignment, or in real time for the environmental or physical change and can help improve or maintain sufficiently high coupling while the optical system is used. The term “sufficiently high” is used here to indicate sufficient coupling to enable the system to function at a performance level (e.g., regarding resolution, accuracy, power consumption, and so forth) for which the system is designed.

(20) Further, the system can help achieve sufficiently high coupling efficiency without directly contacting the end of the optical fiber. Because the system uses contactless alignment of the optical path to the fiber, the system can help reduce contamination, physical wear, and the like of the end of the optical fiber.

(21) An example of an optical system that can incorporate one or more features from the system shown below is an optical fiber-based strain, temperature, or shape sensing system. As a specific example, an optical system couples light into a multi-core optical fiber to sense, in real time or near real-time, a three-dimensional position in space of an element. As another specific example, an optical system couples light into a multi-core optical fiber to sense, in real time or near real-time, a three-dimensional shape of the optical fiber.

(22) Further, an example of an optical system can include a medical or non-medical system. An example of a medical system can include those used for diagnosis or therapy, including surgical systems. In a medical system example, the system described below can be located in an optical path between one or more light sources and one or more cores of a sensing optical fiber, to help

establish and/or maintain sufficiently high coupling efficiency (or coupling efficiencies) of light entering the sensing optical fiber over the course of one or more medical procedures. This is but one example of use for the system described in detail below. Other uses are also possible.
(23) FIG. **1** shows a schematic drawing of an example of an apparatus **1** that includes a system **100** for directing light into an optical fiber. Because the system **100** can identify a feature in an image of the end of the optical fiber, then use the location of the feature to actively align an optical path to a core of the optical fiber, the system can achieve a relatively robust alignment of a light beam to the core of the optical fiber.
(24) A controller **10** can include various optical and electronic components. For medical applications, the controller **10** can be configured as a piece of capital equipment, which can be used and reused for multiple procedures. For applications directed to shape sensing, such as sensing a three-dimensional orientation or shape of an optical fiber, the controller **10** can include a interrogator **12**. The interrogator **12** can direct light into a fiber and analyze light returning from the fiber. The interrogator **12** can use a technique, such as optical frequency domain reflectometry (OFDR), to determine a three-dimensional position of an optical fiber.
(25) In some examples, a portion of the equipment can be configured as a replaceable element, which can be used for a part of a procedure or several procedures, or for the entirety of one or several procedures, then discarded. The replaceable element can include a catheter **14**, which can include a sensing optical fiber **16** that extends along at least part of the length of the catheter **14**. In a medical example, the catheter **14** and sensing optical fiber **16** can be maintained or reprocessed in a clean environment, or in a sterile environment if clinically required, prior to use.
(26) The system **100** described in detail below can use active alignment to optically connect the optical fiber **16** to the controller **10**. When optically connected, the interrogator **12** can direct light into the sensing optical fiber **16** (through the system **100**), receive light reflected from locations along a length of the sensing optical fiber **16** (also through the system **100**), and analyze the reflected light (such as by OFDR) to determine a strain, temperature, or other physical information of the sensing optical fiber **16**. For a shape sensing application, the interrogator **12** is configured to determine a three-dimensional position or shape of the sensing optical fiber **16**. For clarity, the sensing optical fiber **16** will be referred to in the following discussion as the optical fiber **108**. It will be understood that the optical fiber **108** may include the sensing optical fiber **16** or can optionally include a separate portion of fiber coupled to a proximal end of the sensing optical fiber **16**. References below to the optical fiber **108** can include one or both of these cases.
(27) The controller **10** can include a fiber connection **18**, such as a multi-core fiber or a plurality of single-mode fibers, that can provide light as input to the system **100**. The plurality of single-mode fibers can also be referred to as a bundle of single-mode fibers or a fiber bundle in this document, although the plurality of single-mode fibers may be bunched in a bundle, disposed in a linear array, and so forth. The system **100** can direct the light provided by the fiber connection **18** through various elements in the system **100** to couple into the sensing optical fiber **16**. The light reflected from locations along the length of the sensing optical fiber **16** can return into the system **100**, can propagate through the various elements in the system **100**, can propagate through the fiber connection **18**, and can be processed by the interrogator **12** in the controller **10**. The system **100** can direct a portion of the light through various elements onto one or more detectors. The detector or detectors can generate one or more control signals. The system **100** can use the one or more control signals to control one or more actuatable elements in the system, to improve the coupling efficiency for light entering the optical fiber **108**.
(28) The controller **10** can include an electrical connection **20**, which can provide electrical power to the system **100**. A processor **114**, which can be located in the controller **10** or located in the system **100**, can receive the control signals from the one or more detectors in the system **100** and can drive the one or more actuatable elements in the system **100** to improve the coupling efficiency.
(29) During operation, imaging optics in the system **100** can form an image **104** of an end **106** (also

"end-face") of the optical fiber **108**. An actuatable optical element **110**, such as a pivotable mirror, can define an optical path **112** that extends to the actuatable optical element **110** and further extends to the end **106** of the optical fiber **108** when the optical fiber **108** is present. A processor **114** can determine a location in the image **104** of a specified feature, such as a circumferential edge of the end **106** of the optical fiber **108**, in the image **104**. Although the processor **114** is shown as being located with the system **100**, it will be understood that the processor **114** can alternatively be located with the controller **10**. The processor **114** can cause, based on the location of the specified feature in the image **104**, the actuatable optical element **110** to actuate to align the optical path **112** to a core **116** of the optical fiber **108**.
(30) The optical path **112** is a geometrical construct that extends from optical element to optical element between the fiber connection **18** and the optical fiber **108**. Specifically, one end of the optical path is located at the fiber connection **18**, and the other end is located at the optical fiber **108** when the optical fiber **108** is present. The optical path **112** can be bent, translated, rotated, and otherwise aligned by the optical components during operation of the system **100**. During operation of the system **100**, a light beam **120** is directed to propagate along the optical path **112** from optical element to optical element, so that the light beam **120** follows the optical path **112**. It is instructive to clarify that the optical path **112** can be redirected, both when the light beam **120** is present and when the light beam **120** is absent. For configurations in which the optical fiber **108** includes multiple cores, the system **100** can include multiple optical paths **112** that propagate toward respective cores of the optical fiber **108**.
(31) A light source **118** can direct the light beam **120** along the optical path **112** to couple into a core **116** of the optical fiber **108**. In some examples, the controller **10** can include one or more light-producing elements, such as light-emitting diodes or laser diodes, and one or more light coupling elements, such as lenses, that can direct light from the light-producing elements into one or more cores of the one or more fibers in the fiber connection **18**. In the configuration of FIG. **1**, the light source **118** can include a distal end of the fiber connection **18** or a length of fiber that is coupled to a distal end of the fiber connection **18**. For clarity, the fiber in the fiber connection **18** will be referred to in the following discussion as the source optical fiber **138** (herein also sometimes referred to as "capital-side optical fiber"). It will be understood that the source optical fiber **138** may be the same as the fiber connection **18** or can optionally include a separate portion of fiber coupled to a distal end of the fiber connection **18**.
(32) For examples in which the optical fiber **108** includes a single core **116**, the source optical fiber **138** can include a single core **140**. For examples in which the optical fiber **108** includes multiple cores **116**, the source optical fiber **138** can also include multiple cores **140**. The multiple cores **140** can be arranged in a pattern that resembles the pattern of the multiple cores of the optical fiber **108**. As a specific example, the source optical fiber **138** and the optical fiber **108** can each include six cores located in a hexagonal pattern that surrounds the center of the circumferential edge of the fiber. During operation, the system **100** can simultaneously direct light from the multiple cores **140** of the source optical fiber **138** into the multiple cores of the optical fiber **108**.
(33) For examples in which the optical fiber **108** includes multiple cores **116**, an alternative to receiving light from multiple cores of a multi-core fiber in the fiber connection **18** is receiving light from the cores of a plurality of single-core fibers, such as fibers in a fiber bundle or a linear array of fibers. In some examples, the optical fiber **108** can be a multi-core optical fiber. The core **116** of the optical fiber **108** can be a first core of a plurality of cores of the multi-core optical fiber. The optical path **112** can be a first optical path of a plurality of optical paths defined by the actuatable optical element **110**. Each optical path of the plurality of optical paths can extend to the actuatable optical element **110** and can further extend to the end **106** of the multi-core optical fiber. The processor **114** can cause the actuatable optical element **110** to actuate to align the optical path **112** to the core **116** by causing the actuatable optical element **110** to actuate to simultaneously align the plurality of optical paths to the plurality of cores of the multi-core optical fiber. The light source

118 can be a first light source of a plurality of light sources. Each light source of the plurality of light sources can direct a corresponding light beam along a corresponding optical path of the plurality of optical paths to couple into a corresponding core of the plurality of cores of the multi-core optical fiber.

(34) FIG. **2** shows an end-on view of an example of a light source **118** configured as a fiber bundle **200**, which is suitable for use in the system **100** of FIG. **1**. The fiber bundle **200** includes a plurality of single-mode fibers **202**, **204**, **206**, **208**, **210**, and **212**. These single-mode fibers of the plurality include corresponding cores **216**, **218**, **220**, **222**, **224**, and **226**, respectively. The single-mode fibers of the plurality surround a central fiber **214** having a core **228**. In this example, the single-mode fibers **202**, **204**, **206**, **208**, **210**, and **212** of the plurality are arranged in a pattern of a regular hexagon surrounding the central fiber **214**. The fiber bundle **200** is suitable for use as a light source **118** for an optical fiber **108** having multiple cores arranged in a similarly shaped hexagonal pattern. The fiber bundle **200** of FIG. **2** is but one example of a fiber bundle; other arrangements of fibers are also possible. The system **100** can further include magnification optics that can impart a magnification to the plurality of optical paths. The magnification can equal, or substantially equal to within a tolerance, such as 1%, 5%, 10%, or 20%, a ratio of a spacing between adjacent cores of the plurality of cores of the multi-core optical fiber to a spacing between adjacent cores of the plurality of single-core fibers. The magnification optics can include a source objective element **148** (described in detail below), which can collimate light emerging from the light source **118** to form the light beam **120**, and an objective element **122** (also described in detail below), which can focus the light beam **120** to couple into the optical fiber **108**. The ratio of the focal lengths of the source objective element **148** and the objective element **122** can be selected to equal, or substantially equal the ratio of the spacing between adjacent cores of the plurality of cores of the multi-core optical fiber of the optical fiber **108** to a spacing between adjacent cores of the plurality of single-core fibers of the light source **118**.

(35) Returning to FIG. **1**, in some examples, the processor **114** can cause the actuatable optical element **110** to actuate to align the optical path **112** to the core **116** by using at least the following two operations. First, the processor **114** can determine an offset between the location of the specified feature in the image **104** and a predetermined target location in the image **104**. Second, the processor **114** can cause the actuatable optical element **110** to actuate to reduce the offset. The processor **114** can optionally repeat these two operations during operation of the system **100** to help maintain a sufficiently high coupling efficiency into the core **116** during operation. For example, the processor **114** can determine a pixel location (e.g. a set of orthogonal location coordinates, such as x and y) in the image **104** of the specified feature (such as a center of a circumference of the optical fiber **108**), can compare the determined pixel location to a specified pixel location (e.g., such as a set of values saved in a lookup table or other suitable memory) that corresponds to a well-aligned optical fiber **108**, and cause the actuatable optical element **110** to actuate to move the determined pixel location to coincide with the specified pixel location.

(36) In some examples, the specified feature can include part or all of a circumferential edge of the end **106** of the optical fiber **108**. The core **116** can be located at a predetermined core location relative to the circumferential edge of the optical fiber **108**. The processor **114** can cause the actuatable optical element **110** to actuate to align the optical path **112** to the core **116** by causing alignment of the optical path **112** to the predetermined core location. For example, for configurations in which the optical fiber **108** is a single-core fiber, the core **116** can be located at a center of the circumferential edge of the optical fiber **108**. For configurations in which the optical fiber **108** is a multi-core fiber (e.g., a fiber in which a single cladding surrounds multiple cores that are spaced apart from one another), the cores **116** can be located at specified locations with respect to the circumferential edge of the optical fiber **108**. For example, the optical fiber **108** can include four cores, with a center core located at a center of the circumferential edge and three cores located at corners of an equilateral triangle centered about the center core. As another example, the optical

fiber **108** can include six cores located in a hexagonal pattern that surrounds the center of the circumferential edge of the optical fiber **108** (e.g., the cores can be located at the corners of a regular hexagon by being spaced apart azimuthally by sixty degrees, about sixty degrees, or sixty degrees to within a tolerance of one degree, two degrees, five degrees, or another suitable value). As yet another example, the optical fiber **108** can include seven cores, with a center core located at a center of the circumferential edge and six cores located at corners of a regular hexagon centered about the center core. Other suitable multi-core configurations can also be used.

(37) For configurations in which the optical fiber **108** includes multiple cores, the circumference of the optical fiber **108** can include an optional azimuthal locating feature, such as a partially flattened edge, a notch, a protrusion, or other feature that can mechanically or optically indicate the azimuthal locations of the cores. For example, the optical fiber **108** can include a rod (not a core) that extends along a length of the optical fiber **108**. Such rod can appear as a bright dot (e.g., brighter than an area surrounding the dot) or a dark dot (e.g., darker than an area surrounding the dot) in the image **104** of the end **106** of the optical fiber **108**. In some examples, the azimuthal locating feature can be a mechanical feature of a connecting element. For example, the optical fiber **108** can be held in a ferrule and standard optical connector. When the connector is manufactured, a specific core of the optical fiber **108** can be illuminated, to align the specific core to a key of the standard optical connector.

(38) Other specified features can also be used in addition to or instead of the circumferential edge of the end **106** of the optical fiber **108**. For example, the feature can include the appearance of the core **116** of the optical fiber **108** in the image **104**. In some illumination configurations, the core **116** can appear as a dark spot in the image **104**, which can appear darker (e.g., with a lower intensity or brightness) than an area surrounding the core **116** (see, e.g., FIG. **13** below). In other illumination configurations, the core **116** can appear as a bright spot in the image **104**, which can appear brighter (e.g., with a higher intensity or brightness) than an area surrounding the core **116**. Identifying the core **116** directly from bright spots and/or dark spots in the image can also be used with multi-core fibers that have multiple cores.

(39) The system **100** can optionally further include an illumination light source **130**. The illumination light source **130** can illuminate the optical fiber **108** with illumination **132**. The illumination **132** can have a wavelength different from a wavelength of the light beam **120**. As a specific example, the wavelength of the light can be 1550 nm, and the wavelength of the illumination **132** can be in the visible spectrum, such as between 400 nm and 700 nm. Other wavelength values can also be used.

(40) For configurations that include the illumination light source **130**, at least some of the illumination **132** can reflect or scatter from the optical fiber **108** to form first light. In some examples, illumination **132** that reflects off the end **106** of the optical fiber **108** can produce the first light. In some examples, illumination that enters a side of the optical fiber **108** and exits the end **106** of the optical fiber **108** can form the first light.

(41) An objective element **122** can collimate at least some of the first light to form second light. In some examples, such as the configuration of FIG. **1**, the objective element **122** can include an objective lens. The optical path **112** can extend through the objective lens. As an alternative, the objective element **122** can include an objective mirror. FIG. **3** shows a top view of an example of a portion of the system **100** of FIG. **1**, in which the objective element **122** is configured as an objective mirror **302**. The objective mirror **302** can have a cross-section that includes a section of a parabola. Other configurations for the objective element **122** are also possible, including multiple mirrors, multiple lenses, or a combination of at least one mirror and at least one lens. Similarly, the focusing element **126** can include at least one of a focusing lens or a focusing mirror. FIG. **11** is a simplified diagram showing selective components of the system of FIG. **1**, where the objective element **122**, source objective element **148**, and focusing element **126** are parabolic mirrors rather than lenses. Glass lenses generally have chromatic aberrations that change the location of the in-

focus image depending upon wavelength. For that reason, either a specially designed achromatic lens or a parabolic mirror provides a better solution. Beneficially, parabolic mirrors are readily available.

(42) Returning to FIG. **1**, a dichroic mirror **124** can direct at least some of the second light away from the optical path **112** to form third light. For example, the dichroic mirror **124** can transmit light in a transmission band that includes 1550 nm. The dichroic mirror **124** can reflect light in a reflection band that includes the wavelength of the illumination **132**, such as in the visible spectrum. This is but one numerical example; other wavelengths and wavelength ranges can also be used.

(43) In the configuration of FIG. **1**, the dichroic mirror **124** is a long pass dichroic mirror, which can transmit relatively long wavelengths (such as those used for performing the shape sensing, optionally in the infrared portion of the electromagnetic spectrum such as 1550 nm), and reflect relatively short wavelengths (such as those used for performing the imaging functions, optionally in the visible portion of the electromagnetic spectrum such as between 400 nm and 700 nm). Alternatively, the dichroic mirror **124** can be a short pass dichroic mirror, which can transmit the relatively short wavelengths (such as those used for performing the imaging functions) and reflect the relatively long wavelengths (such as those used for performing the shape sensing). Replacing the long pass filter with a short pass dichroic mirror would involve swapping the transmitted and reflected arms, so that the optical path **112** would reflect at the dichroic mirror **124** rather than transmit through the dichroic mirror **124** as currently shown in FIG. **1**.

(44) A focusing element **126** can focus the third light to form the image **104** at a focal plane of the focusing element **126**. An imaging array **128** can be located at the focal plane of the focusing element **126** and can sense the image **104**. In some examples, the imaging optics can include the objective element **122**, the dichroic mirror **124**, the focusing element **126**, and the imaging array **128**. The processor **114** can receive from the imaging array **128** an analog and/or a digital signal that corresponds to the image **104**. Other suitable configurations can also be used.

(45) In the example of FIG. **1**, the actuatable optical element **110** is configured as a pivotable mirror. The pivotable mirror can include a single mirror that can pivot in two dimensions, two separated mirrors that can each pivot in a single dimension, multiple mirrors that can each pivot in a single dimension or two dimensions, and other suitable configurations. In the configuration of FIG. **1**, the pivotable mirror can include a reflective mirror that can pivot about a pivot point, and a linear actuator **136** that can pivot the reflective mirror about the pivot point. Although the pivotable mirror is shown in FIG. **1** as pivoting in only one dimension, it will be understood that the pivotable mirror can pivot in two orthogonal dimensions, using a pair of linear actuators **136**. The processor **114** can control the linear actuators **136**. The processor **114** can actuate the actuatable optical element **110** to align the optical path **112** to the core **116** of the optical fiber **108** by pivoting the pivotable mirror to steer the optical path **112** based on the location of the specified feature in the image **104**.

(46) The optical path **112** can include a fixed portion, extending between the light source **118** and the actuatable optical element **110**. The optical path **112** can include a movable portion, extending between the actuatable optical element **110** and the end **106** of the optical fiber **108**. During operation of the system **100**, the movable portion of the optical path **112** can move in space, while the fixed portion of the optical path **112** may remain stationary. In the configuration of FIG. **1**, the dichroic mirror **124**, focusing element **126**, and imaging array **128** are located in the fixed portion of the optical path **112**. Other configurations can also be used.

(47) In some examples, the actuatable optical element **110** can be located in the optical path **112** to be telecentric. For a telecentric configuration, pivoting the pivotable mirror can produce lateral translation of the optical path **112** at the end **106** of the optical fiber **108** without producing a change in angle of the optical path **112** at the end **106** of the optical fiber **108**. In some examples, locating the pivotable mirror at a rear focal plane (or a back focal plane) of the objective element

122 can produce the telecentric condition.
(48) The pivotable mirror of FIG. **1** is but one example of a suitable actuatable optical element **110**. Other suitable configurations can include a translatable optical element, such as a translatable lens or a translatable mirror. In some examples, the translatable optical element can include the objective element **122**, the full system **100**, and/or the optical fiber **108**.
(49) In some examples, the system **100** can include features that allow the system **100** to operate in a separate environment, such as a clean-room environment in an industrial example, or a sterile environment in a medical example involving sterility. For example, in some applications in which a medical procedure is performed, such as when the system **100** can be reusable (e.g., can be capital equipment), and the optical fiber **108** can be replaceable (e.g. can be disposed of after a single-use, or reprocessed and disposed of after multiple uses, or be reprocessed for an indefinite number of uses), the system **100** can optionally include a barrier, such as a window or optical surface. In medical examples, the barrier may meet cleanliness requirements to help provide a clean environment for particular medical procedures not requiring sterility or may meet sterility requirements to help ensure sterility for medical procedures requiring sterility.
(50) The window or optical surface can pass the light beam **120** to the optical fiber **108** and can receive light from the optical fiber **108**, without contacting the optical fiber **108**. In some examples, the window or optical surface, can be easily cleaned between uses of the system **100**, to avoid contaminating optical fibers used in subsequent procedures. In some examples, the objective element **122**, such as the objective lens, can form part of a barrier for the system **100**. For example, the objective lens can be plano-convex, with the planar side optionally forming part of the sterile barrier. Other configurations can also be used. As noted above, in some examples, the barrier formed by the system **100** may not be a sterile barrier in that it does not meet sterility requirements.
(51) The system **100** can optionally further include a field aligning lens **134** located in the optical path **112** proximate the end **106** of the optical fiber **108**. Such a field aligning lens **134** can improve the coupling efficiency for cases when the optical fiber **108** is positioned away from a central axis of the optical elements of the system **100** (e.g., off-axis performance). The field aligning lens **134** can optionally have a same focal length as the objective element **122**. The field aligning lens **134** can optionally have a diameter (e.g. a clear aperture) than is less than a diameter of the objective element **122**. The field aligning lens **134** can optionally have a numerical aperture (e.g., half the diameter, divided by the focal length) than is less than a numerical aperture of the objective element **122**. The field aligning lens **134** can optionally be formed as a plano-convex lens. The field aligning lens **134** can optionally have a planar side that forms part of a sterile barrier of the system **100**. Because the field aligning lens **134** may be a relatively inexpensive item, the field aligning lens **134** can optionally be configured as a replaceable (e.g., single-use or multi-use) element that can be removed, reprocessed, reused, and/or disposed of. Such a replaceable element can optionally be packaged with, or separately from, the optical fiber **108**.
(52) In some examples, the system **100** can optionally monitor an amount of light that is reflected from one or more cores of the optical fiber **108**. For example, in a position-sensing application, the system **100** can couple light into one or more cores of the optical fiber **108**, light can reflect in varying amounts from locations along a length of the optical fiber **108**, and the system **100** can analyze the reflected light, such as by optical frequency domain reflectometry (OFDR) performed by the interrogator **12**, to determine a three-dimensional position of the optical fiber **108**. In some examples, the analysis of the reflected light can include sensing a magnitude or amplitude of the reflected light. Such a sensed magnitude or amplitude can correspond to a coupling efficiency of light entering the optical fiber **108**. The system **100** can actuate the actuatable optical element **110** to raise, maximize, and/or optimize the sensed magnitude or amplitude of the reflected light from the optical fiber **108**.
(53) In some examples, the system **100** can use the sensed magnitude or amplitude in concert with the imaging technique described above. For example, the system **100** can use the imaging

technique to perform an initial positioning of the optical path **112** near or at the core **116** (e.g., as a coarse alignment procedure), and can use the sensed magnitude or amplitude to more precisely position the optical path **112** with respect to the core **116** (e.g., as a fine alignment procedure). In some examples, the system **100** can use the sensed magnitude or amplitude to position the optical path **112** with respect to the core **116**, without using the imaging technique described above. In some examples, the system **100** can use the sensed magnitude or amplitude to register the camera image acquired by the imaging array **128** to the settings of the actuatable optical element **110** (e.g., implemented by a steering mirror). The steering mirror can be scanned, and the back-reflected light coupled into the capital-side fiber (corresponding to source optical fiber **138**) can be plotted as a function of mirror position. Scanning the mirror can generate an image of the sensor end-face **106** similar to the image gathered (by the imaging array **128**) using visible light and standard imaging. In this case, the cores are bright because of reflection from the gratings in the cores. Images such as this can take significant time to gather: a 1024×1024 pixel scan, for example, may take 26 minutes. This delay may be far too long to use in a surgical setting. It can be used, however, to register the camera image of the sensor end-face **106** to the steering mirror settings. This calibration then allows the mirror settings to be determined by the location of the cores in the visible-light image.

(54) As explained above, the system **100** can illuminate the optical fiber **108** to capture the image **104** of the end **106** of the optical fiber **108**. For configurations in which the light source **118** includes a source optical fiber **138**, the system **100** can optionally illuminate an end of the source optical fiber **138** and include additional optical elements to superimpose a view of an end **142** of the source optical fiber **138** onto the view of the end **106** of the optical fiber **108** in the image **104**. Allowing the ends of the two fibers to be viewed simultaneously can provide additional information during the assembly and alignment stages of the system **100**.

(55) As explained above, a first illumination light source, such as **130**, can illuminate the optical fiber **108** with first illumination, such as **132**. The first illumination **132** can have a first wavelength different from a wavelength of the light beam **120**. At least some of the first illumination **132** can reflect or scatter from the optical fiber **108** to form first light. An objective element **122**, such as an objective lens or objective mirror, can collimate at least some of the first light to form second light. A second illumination light source **144** can illuminate the source optical fiber **138** with second illumination **146**. The second illumination **146** can have a second wavelength that is different from the first wavelength and different from the wavelength of the light beam **120**. At least some of the second illumination **146** can reflect or scatter from the source optical fiber **138** to form third light. A source objective element **148**, such as a source objective lens or source objective mirror, can collimate at least some of the third light to form fourth light. A dichroic mirror, such as **124**, and a reflector **150**, such as a retroreflector or retroreflecting prism, can superimpose the second light and the fourth light to form fifth light. In the configuration shown in FIG. **1**, the dichroic mirror **124** can reflect at least some of the fourth light toward the reflector **150**. Alternatively, the dichroic mirror **124** can be oriented to reflect at least some of the second light toward the reflector **150**. A focusing element, such as **126**, can focus the fifth light to form the image **104** at a focal plane of the focusing element **126**. An imaging array, such as **128**, located at the focal plane of the focusing element **126**, can sense the image **104**. Because the ends **106**, **142** of the fibers can be imaged with light at different wavelengths, the processor **114** can optionally separate the information from the two superimposed views as needed. Forming the superimposed views of the ends **106**, **142** of the fibers can provide additional information during the assembly and/or alignment stages of the system **100**, and/or during use of the system **100**. For example, because the ends of the fiber can be visible in the image **104**, the image **104** can be used to check the fiber ends for contamination or damage.

(56) As an alternative to illuminating the end **142** of the source optical fiber **138**, or in addition to performing such illumination, the controller **10** can inject illumination into an opposite end of the source optical fiber **138**, which can propagate along the source optical fiber **138** to emerge from the end **142** of the source optical fiber **138**. Because the illumination for imaging can use a different

wavelength than the light used for shape sensing, the injection of the illumination can be performed by wavelength division multiplexing at the controller **10**. FIG. **12** shows a schematic diagram of an example system (corresponding to a particular embodiment of system **100**) that allows imaging of both the capital-side (or source) optical fiber **138** and the multi-core sensor (corresponding to optical fiber **108**), and thus determining the relative position of the images. Rapid and continuous measurement of the position of the image of the capital-side cores with respect to the cores of the sensor can be achieved by injecting visible light into one or more of the capital-side cores. Various existing fibers include an extra core (beyond the cores used for shape sensing) that may be used for this purpose. Alternatively, one of the cores used for shape sensing (e.g., core **190** in FIG. **12**) may be used by injecting visible light (e.g., from a visible light LED **192**) using a wavelength division multiplexer. The reflection of the capital-side light off of the sensor fiber interface can then be imaged. The image generated by this type of system is shown in FIG. **13**. The sensor cores **1300** (only two labeled) appear dark, and the image of the two illuminated capital-side cores **1302** are bright. With the image of the sensing fiber and the image of at least one of the cores of the capital-side fiber in the same image, it is straightforward to steer the mirror (or other actuatable optical element **110**) such that the core images are superimposed and good coupling is achieved.

(57) With renewed reference to FIG. **1**, the optical path **112** can include an optional first pivotable element **152** that can redirect the optical path **112** within an angular range that extends in one dimension or in two dimensions. The first pivotable element **152** can include a mirror on adjustable mount that can controllably pivot about one, two, three, or more axes. Where the pivotable element be pivoted about multiple axes, the axes may intersect or not intersect, or be orthogonal to each other or be rotationally offset by some other angle. The phrase “pivotable element” is intended to include a variety of “tip/tilt elements” that can include, for example, elements such as mirrors, mounted on a tip/tilt stage. A tip/tilt stage can typically pivot about each of two orthogonal and intersecting axes, although other configurations can also be used. In the configuration of FIG. **1**, the first pivotable element **152** can have a nominal angle of incidence of 45 degrees, or about 45 degrees, so that the optical path **112** can be nominally redirected by 90 degrees, or about 90 degrees. The incidence angle of 45 degrees is but one example of an incident angle; other suitable angles of incidence can also be used. The first pivotable element **152** can provide an additional degree of freedom during the assembly and alignment stages of the system **100**. For example, locating the first pivotable element **152** in the optical path **112** can help relax some placement tolerances on the source optical fiber **138**, and can help compensate for rotations and/or displacements of other optical elements in the optical path **112**. The first pivotable element **152** can be located in the optical path **112** between the light source **118** and the dichroic mirror **124**, between the dichroic mirror **124** and the actuatable optical element **110**, between the actuatable optical element **110** and the optical fiber **108**, or at any other suitable location along the optical path **112**.

(58) The optical path **112** can include an optional second pivotable element **154** that can redirect the optical path **112** within an angular range that extends in one dimension or in two dimensions. The second pivotable element **154** can be similar in structure and function to the first pivotable element **152**. The second pivotable element **154** can be located in the optical path **112** between the light source **118** and the dichroic mirror **124**, between the dichroic mirror **124** and the actuatable optical element **110**, between the actuatable optical element **110** and the optical fiber **108**, or at any other suitable location along the optical path **112**. The first pivotable element **152** and the second pivotable element **154** can be located at different locations along the optical path **112** (e.g., can be longitudinally separated along the optical path **112**). Although the second pivotable element **154** is shown in FIG. **1** as being adjacent to the first pivotable element **152** with no intervening optical elements between them, the second pivotable element **154** can be located at any suitable location along the optical path **112**, including between beamsplitters, or between a beamsplitter and the actuatable optical element **110**. Using two pivotable elements that are separated along the optical

path **112** can be helpful during the assembly and alignment of the optical components in the system **100**. For example, using two longitudinally separated pivotable elements can allow the optical path **112** to be laterally translated (e.g., moved without rotation) to a desired location, or rotated in two dimensions while keeping a fixed spatial location. As a specific example, using two pivotable elements can allow the optical path **112** to pass through a center of a lens, rather than the edge of a lens, to improve the optical performance of the lens.

(59) FIG. **14** shows a schematic diagram of an example system (corresponding to a particular embodiment of the system **100**) for making a non-contact connection between a capital-side fiber (serving as the light source **118**) and a multi-core sensor (corresponding to optical fiber **108**) that uses two tip/tilt mirrors (e.g., implementing pivotable elements **152**, **154**) for beam steering and alignment. If a tip/tilt mirror is used to steer the beam to compensate for connector x/y misalignments of more than 1-200 microns, significant distortions can occur because the light passes through the side of the lens (or other objective element **122**) rather than the center. This leads to relatively large losses for larger sensor displacements. To compensate for this, two tip/tilt mirrors (**152**, **154**) can be used to steer the beam. In this configuration, the beam can be steered to be at the right angle to the lens (or other objective element **122**) to enable coupling to an offset sensor (corresponding to optical fiber **108**) while still passing through the center of the lens.

(60) FIGS. **15** and **16** show a simple model of a system with one and two mirrors, respectively. Both systems are set up to focus light on a fiber displaced by 300 microns from the optimal optical axis of the system, which is centered on the lens (or other objective element **122**). In the first model (FIG. **15**), with one mirror (e.g., implementing pivotable element **152**), one can see that the beam is well off-center of the lens. In this model, coupling losses were about 1.96 dB one way. In the second model (FIG. **16**), the two mirrors (e.g., implementing pivotable elements **152**, **154**) are used to steer the beam to be more centered on the lens while still coupling to the offset fiber core. This reduces the one-way loss to 0.92 dB, which is a significant improvement.

(61) In some examples, it may be beneficial to use one or more pivotable elements to steer the optical path **112** such that the optical fiber **108** and the source optical fiber **138** can be approximately parallel to each other (at least to within a few degrees). Orienting the fibers to be approximately parallel is generally consistent with use of typical physical contact connectors, which typically require insertion of the fibers from opposing sides. The configuration of FIG. **1** can be modified to achieve this parallelism condition by removing one of the pivotable elements **152** or **154**, adding an additional pivotable element, modifying the dichroic mirror **124** to be a short pass filter rather than a long pass filter, and other geometrical modifications.

(62) Thus far, discussion has focused on laterally aligning the optical path **112** to a core **116** of the optical fiber **108**. Specifically, for a coordinate system (x, y, z) at the end **106** of the optical fiber **108**, in which z corresponds to a central axis of the optical path **112**, the discussion above is directed to aligning the optical path **112** in the x- and y-dimensions. For example, capturing an image **104** of the end **106** of the optical fiber **108** can provide x- and y-coordinates of one or more features on the optical fiber **108**, and the system **100** can actively align the optical path **112** in x- and y-dimensions with respect to the feature or features in the image **104**.

(63) In some applications, active alignment in x- and y-dimensions may be sufficient to achieve sufficiently high coupling into the optical fiber **108**. These applications can rely on mechanical placement of the end **106** of the optical fiber **108** as being sufficiently accurate to achieve the sufficiently high coupling. For example, the optical fiber **108** may fit into a clamp that can position the end **106** of the optical fiber **108** in a specified plane (in the z-direction) to within a specified tolerance, such that for any z-position within the tolerance, the coupling efficiency is sufficiently high.

(64) In other applications, the mechanical placement in the z-direction may not be suitably accurate to achieve the sufficiently high coupling. For these applications, the system **100** can further include one or more longitudinal position sensor elements. The longitudinal position sensor elements can

detect a longitudinal separation (e.g., a distance as measured along the optical path **112**) between a focus of the light beam **120** and the end **106** of the optical fiber **108**. The longitudinal position sensor elements can be located at any suitable location in the fixed portion of the optical path **112**.
(65) Similarly, the system **100** can further include one or more longitudinal position adjustor elements. The longitudinal position adjustor element or elements can longitudinally position the focus of the light beam **120** to reduce the longitudinal separation between the focus and the **106** of the optical fiber **108**. The longitudinal position adjustor element or elements can be located at any suitable location in the fixed portion of the optical path **112**. In some examples, the longitudinal position adjustor element or elements can be located in the optical path **112** between the longitudinal position sensor elements and the optical fiber **108**.
(66) In the configuration of FIG. **1**, the longitudinal position sensor elements can include a beamsplitter **156**, such as a dichroic beamsplitter, a 50-50 beamsplitter, or another suitable beam-splitting element. In the configuration of FIG. **1**, the beamsplitter **156** can be located between the dichroic mirror **124** and the optical fiber **108** along the optical path **112**. Alternatively, the dichroic mirror **124** can be located between the beamsplitter **156** and the optical fiber **108** along the optical path **112**. Other configurations can also be used, including configurations that can swap the functions of the transmitted and reflected paths through the beamsplitter **156**.
(67) The beamsplitter **156** can receive light that has been reflected and/or scattered from the end **106** of the optical fiber **108**. The beamsplitter **156** can direct a portion of the reflected light toward a bi-prism **158**, a lens **160**, and a sensor **162**. The lens **160** can focus the light emergent from the bi-prism **158** to form an image **164** at the sensor **162**. The sensor **162** can be coupled to the processor **114**.
(68) The bi-prism **158** can impart a wedge angle between opposing halves of the reflected light such that the specified feature, such as the circumferential edge of the optical fiber **108**) in the image **164** has a corresponding duplicate feature in the image **164**. The processor **114** can further determine, based at least in part on a spacing between the specified feature and the corresponding duplicate feature, the longitudinal separation between the focus and the end **106** of the optical fiber **108**. For these configurations, the spacing can be considered to be a focus error signal. The spacing can be compared against a specified distance, which can be determined in an initial configuration of the system **100**, such as at a factory during initial assembly and alignment of the system **100**. If the spacing is less than the specified distance, the focus can be on one side of the end **106** of the optical fiber **108**, such as outside the optical fiber **108**. If the spacing is greater than the specified distance, the focus can be on the other side of the end **106** of the optical fiber **108**, such as within the optical fiber **108**.
(69) The bi-prism configuration of FIG. **1** is but one example of a configuration for the longitudinal position sensor elements. Other suitable configurations are shown in FIGS. **4-6** and described below.
(70) FIG. **4** shows a top view of an example of longitudinal position sensor elements that are suitable for use in the system **100** of FIG. **1**. The beamsplitter **156** in the optical path **112** directs a light portion **402** of the light beam **120** that has been reflected and/or scattered from the end **106** (FIG. **1**) of the optical fiber **108** (FIG. **1**) toward a split-field dichroic filter **404**, a lens **406**, and a sensor **410**. A first half **404**A of the split-field dichroic filter **404** can have a first spectral profile (e.g., can pass a first wavelength or a first wavelength band). A second half **404**B of the split-field dichroic filter **404** can have a second spectral profile different from the first spectral profile (e.g., can pass a second wavelength different from the first wavelength or a second wavelength band different from the first wavelength band). The light portion **402** can form an image **408** at the sensor **410**. The sensor **410** can be coupled to the processor **114**.
(71) The light portion **402** can have a plurality of wavelengths. The split-field dichroic filter **404** can be configured such that a specified feature in the image **408** has a corresponding duplicate feature in the image **408** at a different wavelength. The processor **114** can further determine, based

at least in part on a spacing between the specified feature and the corresponding duplicate feature, the longitudinal separation between the focus and the end **106** of the optical fiber **108**. The spacing can be compared against a specified distance, as explained above.

(72) FIG. **5** shows a top view of another example of longitudinal position sensor elements that are suitable for use in the system **100** of FIG. **1**. The beamsplitter **156** in the optical path **112** directs a light portion **502** of light beam **120** that has been reflected and/or scattered from the end **106** (FIG. **1**) of the optical fiber **108** (FIG. **1**) toward a liquid crystal on silicon (LCOS) device **504** that has a programmable aperture, a lens **506**, and a sensor **510**. The light portion **502** can form an image **508** at the sensor **510**. The sensor **510** can be coupled to the processor **114**.

(73) The LCOS device **504** can use time multiplexing to achieve a similar splitting effect achieved by the elements shown in FIGS. **1** and **4**. At a first time, a first half **504**A of the aperture of the LCOS device **504** can be reflective, while a second half **504**B of the aperture of the LCOS device **504** can be non-reflective. During the first time, the processor **114** can acquire a first image from the sensor **510**. At a second time after the first time, the first half **504**A of the aperture of the LCOS device **504** can be non-reflective, while the second half **504**B of the aperture of the LCOS device **504** can be reflective. During the second time, the processor **114** can acquire a second image from the sensor **510**. The LCOS device **504** can be configured such that a specified feature in the first image has a corresponding duplicate feature in the second image. The processor **114** can further determine, based at least in part on a spacing between the specified feature and the corresponding duplicate feature, the longitudinal separation between the focus and the end **106** of the optical fiber **108**. The spacing can be compared against a specified distance, as explained above.

(74) FIG. **6** shows a top view of another example of longitudinal position sensor elements that are suitable for use in the system **100** of FIG. **1**. The beamsplitter **156** in the optical path **112** directs a light portion **602** of light beam **120** that has been reflected and/or scattered from the end **106** (FIG. **1**) of the optical fiber **108** (FIG. **1**) toward a chromatically aberrated lens **606** and a sensor **610**. The light portion **602** can form an image **608** at the sensor **610**. The sensor **610** can be coupled to the processor **114**.

(75) The light portion **602** can have a plurality of wavelengths. Because the chromatically aberrated lens **606** includes chromatic aberration, the chromatically aberrated lens **606** can bring light at one wavelength to a first focus at a first focal plane, and can bring light at a second wavelength (the second wavelength different from the first wavelength) to a second focus at a second focal plane that is separated from the first focal plane. Note that in the absence of chromatic aberration, which is typically the case with most well-designed lenses that operate at more than one wavelength, the first and second focal plane are often coincident or are nearly coincident.

(76) The chromatically aberrated lens **606** can be configured such that a specified feature in the image **608** has a corresponding duplicate feature in the image **608** at a different wavelength. The processor **114** can further determine, at least in part from a size of the specified feature in the image **608** and a size of the corresponding duplicate feature in the image **608**, the longitudinal separation between the focus and the end **106** of the optical fiber **108**. In addition, or as an alternative, the processor **114** can process the image **608**, such as by perform a two-dimensional Fast Fourier Transform on the image **608**, to evaluate the sharpness of the image **608** at different wavelengths. The sharpness at the different wavelengths can help determine the longitudinal separation between the focus and the end **106** of the optical fiber **108**, and/or can help determine at least a sign of the longitudinal separation (e.g. positive or negative).

(77) The configuration of FIG. **1** uses distinct beamsplitters (such as dichroic mirror **124** and beamsplitter **156**), distinct focusing elements (such as focusing element **126** and lens **160**), and distinct sensors (such as imaging array **128** and sensor **162**) to perform the tasks of imaging (such as using dichroic mirror **124**, focusing element **126**, imaging array **128**) and of focus sensing (such as using beamsplitter **156**, lens **160**, and sensor **162**). As an alternative, the tasks and elements can be combined, such as by using a single beamsplitter (such as dichroic mirror **124**) and moving the

longitudinal position sensing elements (such as the bi-prism of FIG. **1**, the split-field dichroic filter of FIG. **4**, the LCOS device of FIG. **5**, or the chromatically aberrated lens of FIG. **6**) to be located between the focusing element **126** and the imaging array **128**. Such an alternative can be especially effective if the end **106** of the optical fiber **108** is illuminated with multiple wavelengths, such as three wavelengths that can all be located in a reflection band of the dichroic mirror **124** (or a transmission band if the long pass filter of FIG. **1** is replaced with a short pass filter).

(78) FIG. **7** shows a top view of the longitudinal position adjustor elements in the system **100** of FIG. **1**. In FIGS. **1** and **7**, the longitudinal position adjustor element can include a variable focus lens **166**. The variable focus lens **166** can be disposed in the optical path **112**, such as in the fixed portion of the optical path **112**. The processor **114** can further cause the variable focus lens **166** to adjust, based on the longitudinal separation (determined by the longitudinal position sensor elements) between the focus and the end **106** of the optical fiber **108**, a collimation of the light beam **120** to position the focus at the end **106** of the optical fiber **108**.

(79) FIG. **8** shows a top view of an example of longitudinal position adjustor elements suitable for use in the system **100** of FIG. **1**. As an alternative to using the variable focus lens **166** of FIGS. **1** and **7**, the system **100** can include a linear actuator **802** to longitudinally position the objective element **122** with respect to the end **106** of the optical fiber **108**, to longitudinally position the entire system **100** with respect to the end **106** of the optical fiber **108**, to longitudinally position the optical fiber **108** with respect to the system **100**, or to otherwise controllably vary the separation between the system **100** and the end **106** of the optical fiber **108**. The processor **114** can control the linear actuator **802**, based on the longitudinal separation determined by the longitudinal position sensor elements. The linear actuator **802** can adjust a spacing between the focus and the end **106** of the optical fiber **108**. Other suitable actuators and actuator types can also be used.

(80) Any or all of the longitudinal position sensor techniques (such as those that use the bi-prism of FIG. **1**, the split-field dichroic filter of FIG. **4**, the LCOS device of FIG. **5**, the chromatically aberrated lens of FIG. **6**, or others) can be used with any or all of the longitudinal position adjustor techniques (such as that use the variable focus lens of FIGS. **1** and **7**, the linear actuator of FIG. **8**, or others). Further, any or all of the longitudinal position sensor techniques and any or all of the longitudinal position adjustor techniques can be used with any or all of the configurations for the objective element (such as an objective lens or an objective mirror), any or all of the configurations for the optical fiber (such as single-core or multiple-core), any or all of the configurations for the light source (such as a single-core optical fiber, a multi-core optical fiber, a plurality of single-core fibers, or others), and any or all configurations of the pivotable elements (such as including two, omitting one and including just one, omitting both and not including any, or including more than two).

(81) FIG. **9** shows a flowchart of an example of a method **900** for operating a system to direct light into an optical fiber. The system can include imaging optics, a processor, and an actuatable optical element. The actuatable optical element can define an optical path. The optical path can extend to the actuatable optical element and can further extend to the end of the optical fiber. The method **900** can be executed on the system **100** of FIG. **1**, or on any other suitable system.

(82) At operation **902**, the system can generate, with the imaging optics, an image of an end of the optical fiber.

(83) At operation **904**, the system can determine, with the processor, a location in the image of a specified feature in the image.

(84) At operation **906**, the system can cause, with the processor, the actuatable optical element to actuate to align the optical path to a core of the optical fiber based on the location of the specified feature in the image.

(85) At operation **908**, the system can direct a light beam along the optical path to couple into the core of the optical fiber.

(86) In some examples, the method **900** can optionally further include determining, with the

processor, an offset between the location of the specified feature in the image and a predetermined target location in the image. The method **900** can optionally further include causing, with the processor, the actuatable optical element to actuate to reduce the offset.
(87) In some examples, the specified feature can be a circumferential edge of the end of the optical fiber. The core can be located at a predetermined core location relative to the circumferential edge of the optical fiber. The method **900** can optionally further include causing, with the processor, the actuatable optical element to actuate to align the optical path to the core by causing alignment of the optical path to the predetermined core location.
(88) In some examples, the method **900** can optionally further include illuminating the optical fiber with illumination that has a wavelength different from a wavelength of the light beam.
(89) In some examples, the method **900** can optionally further include reflecting or scattering at least some of the illumination from the optical fiber to form first light. The method **900** can optionally further include collimating, with an objective element of the imaging optics, at least some of the first light to form second light. The method **900** can optionally further include directing, with a dichroic mirror of the imaging optics, at least some of the second light away from the optical path to form third light. The method **900** can optionally further include focusing, with a focusing element of the imaging optics, the third light to form the image at a focal plane of the focusing element. The method **900** can optionally further include sensing, with an imaging array located at the focal plane of the focusing element, the image.
(90) In some examples, the method **900** can optionally further include detecting, by a longitudinal position sensor, a longitudinal separation between a focus and the end of the optical fiber. The method **900** can optionally further include positioning, with a longitudinal position adjustor, the focus to reduce the longitudinal separation.
(91) In some examples, the longitudinal position adjustor can further create a duplicate feature in the image. The method **900** can optionally further include determining, by the processor and based at least in part on a spacing or a size difference between the specified feature and the corresponding duplicate feature, the longitudinal separation between the focus and the end of the optical fiber.
(92) In some examples, the longitudinal position adjustor can include a variable focus lens disposed in the optical path. The method **900** can optionally further include causing, by the processor and based on the longitudinal separation between the focus and the end of the optical fiber, the variable focus lens to adjust a collimation of the light beam to position the focus at the end of the optical fiber.
(93) In some examples, the longitudinal position adjustor can include an actuatable objective lens that can direct the optical path onto the end of the optical fiber. The method **900** can optionally further include causing, by the processor and based on the longitudinal separation between the focus and the end of the optical fiber, the actuatable objective lens to move to position the focus at the end of the optical fiber.
(94) FIG. **10** shows a flowchart of an example of another method **1000** for operating a system to direct light into an optical fiber. The optical fiber can include a core. The system can include imaging optics, a first actuatable optical element, and a second actuatable optical element. The method **1000** can be executed on the system **100** of FIG. **1**, or on any other suitable system.
(95) At operation **1002**, the system can generate, with the imaging optics, a first image of an end of the optical fiber.
(96) At operation **1004**, the system can determine, from the first image, a two-dimensional lateral location of the core on the end of the optical fiber.
(97) At operation **1006**, the system can cause, based on the two-dimensional lateral location, the first actuatable optical element to actuate to laterally align an optical path to the core. The method **1000** can repeat operations **1002** through **1006** as needed until the system can determine that the optical path is sufficiently aligned to the core. When the system has completed operation **1006**, optical path is considered to be laterally aligned to the core (e.g., aligned in an x-y plane that is

orthogonal to the optical path at the end of the optical fiber).
(98) At operation **1008**, the system can generate, with the imaging optics, a second image of the end of the optical fiber.
(99) At operation **1010**, the system can determine, from the second image, a longitudinal location of the core on the end of the optical fiber.
(100) At operation **1012**, the system can cause, based on the longitudinal location, the second actuatable optical element to actuate to bring the optical path to a focus at the end of the optical fiber. The method **1000** can repeat operations **1008** through **1006** as needed until the system can determine that the focus is sufficiently close to the end of the optical fiber. When the system has completed operation **1012**, the focus of the optical path is considered to be longitudinally aligned to the end of the fiber (e.g., aligned in a z-direction that is parallel to the optical path at the end of the optical fiber).
(101) In some examples, the first actuatable optical element can include a pivotable mirror. The method **1000** can optionally further include directing a light beam along the optical path to couple into the core of the optical fiber. The method **1000** can optionally further include, repeatedly performing at least the following three operations. First, the system can cause an angular position of the pivotable mirror to dither in two dimensions. Second, the system can sense an amount of light reflected from the core. In some examples, the second operation can be performed by a interrogator of a controller that is coupled to the system. Third, the system can adjust the angular position of the pivotable mirror to increase the amount of light reflected from the core.
(102) In some examples, at least one of the controller or a processor included with the system can take measurements at a plurality of angular positions of the pivotable mirror (x, y) and/or a plurality of focus positions (z), measure the amount of light reflected from the core at each of the angular positions and/or focus positions, fit one or more curves to the measured amounts of light, and adjust the angular positions of the pivotable mirror and/or the focus adjustment to correspond to a local maximum of the one or more curves.
(103) In some examples, the controller **10** can measure a reflected signal, using OFDR, at each location of the mirror search pattern. Various data processing techniques can be used to evaluate where the best fiber coupling is achieved (e.g., which separation between the focus of the light beam **120** and the end **106** of the optical fiber **108** provides the highest amount of light returning from the optical fiber **108** and therefore provides the highest coupling efficiency into the optical fiber **108**).
(104) In one technique, the controller **10** can sum the reflected amplitude in raw frequency domain data and report the overall reflected amplitude. Such a technique will work, but it is possible to additionally exclude some relatively large background signals. The techniques described below can exclude these relatively large background signals and can therefore increase a signal-to-noise ratio of the coupling efficiency measurement.
(105) In another technique, the controller **10** can select a section of the reflected amplitude in the time domain, or optical delay domain, that represents a section of the optical fiber **108** with gratings in the cores, and sum over the selected section. For example, in the time domain, because the horizontal axis corresponds to round-trip propagation time, reflections arising from optical interfaces (such as an interface between glass and air, such as a face of a lens or window) show up as peaks along the horizontal axis. As such, the controller **10** can effectively ignore the peaks arising from these optical interfaces, and analyze data arising from light that is reflected from locations along the length of the optical fiber **108**. The controller **10** can sum the reflected amplitude for data arising from reflection(s) from along the length of the optical fiber **108**, and exclude data arising from reflection(s) from the end **106** of the optical fiber **108** or from other optical surfaces. Selecting which data to use in this manner can increase sensitivity, compared to using all the light that returns to the controller **10**. For example, selecting which data to use in this manner can show a higher signal (e.g., above a noise level) only when light is coupled into the

optical fiber **108** and is reflecting from the grating structures along the length of the optical fiber **108**.

(106) In still another technique, the controller **10** can select the entire optical fiber **108** region in the time domain in which gratings are present, transform the data into the frequency domain, such as by Fast Fourier Transform, and sum the amplitude only over the spectral region in which the gratings reflect. Processing the amplitude data in this manner can help reduce or eliminate low-level broadband reflected amplitude that can arise from connector reflections and other optical interfaces.

(107) A computer-readable medium can store instructions that, when executed by a processor of a system for directing light into an optical fiber, cause the processor to execute operations. The system can include an actuatable optical element that defines an optical path. The optical path can extend to the actuatable optical element and can further extend to the end of the optical fiber. The operations can include at least the following four operations. First, the system can generate, with imaging optics, an image of an end of the optical fiber. Second, the system can determine, from the image, a location in the image of a specified feature in the image. Third, the system can cause the actuatable optical element to actuate to align the optical path to a core of the optical fiber based on the location of the specified feature in the image. Fourth, the system can direct a light beam along the optical path to couple into the core of the optical fiber.

(108) Although the various aspects of the present invention have been described with respect to a preferred embodiment, it will be understood that the invention is entitled to full protection within the full scope of the appended claims.

Claims

1. A system for directing light emitted by first optical cores of a first light source into second optical cores of an optical fiber sensor, the system comprising: a first objective element and a second objective element for directing the light from the first optical cores along respective optical paths; an actuatable optical element disposed between the first objective element and the second objective element; a second light source configured to direct illumination toward an end of the optical fiber sensor, such that the illumination reflects or scatters from the end as reflected or scattered illumination, wherein the illumination has an illumination wavelength differing from a wavelength of the light emitted by the first optical cores; a focusing element and an imaging array configured to form, from the reflected or scattered illumination, an image of the end; and a control system comprising one or more processors, the control system configured to perform operations comprising: determining a location of a specified feature in the image, and causing, based on the location, an actuation of the actuatable optical element to align the optical paths with the respective second optical cores.
2. The system of claim 1, wherein the first light source comprises: an optical fiber comprising the first optical cores and a cladding surrounding the first optical cores; or a bundle of optical fibers, each optical fiber of the bundle comprising an optical core of the first optical cores.
3. The system of claim 1, further comprising: a physical barrier, the physical barrier comprising a window configured to separate the second objective element from the optical fiber sensor.
4. The system of claim 1, wherein the specified feature comprises an aspect selected from the group consisting of: a circumferential edge of the end of the optical fiber sensor, an azimuthal locating feature on a circumference of the end; and a core of the second optical cores.
5. The system of claim 1, wherein: the first objective element comprises a first objective lens, and the second objective element comprises a second objective lens; or the first objective element comprises a first parabolic mirror, and the second objective element comprises a second parabolic mirror.
6. The system of claim 1, wherein the actuatable optical element comprises a pivotable mirror.

7. The system of claim 6, wherein the pivotable mirror is located telecentrically relative to the second objective element, such that pivoting the pivotable mirror produces lateral translation of the optical paths at the end of the optical fiber sensor without producing a change in angle of the optical paths at the end of the optical fiber sensor.
8. The system of claim 1, wherein the actuatable optical element comprises a first pivotable mirror and a second pivotable mirror, the first and second pivotable mirrors physically configurable to together steer the light from the first optical cores towards a center of the second objective element.
9. The system of claim 1, wherein causing the actuation of the actuatable optical element comprises: determining an offset between the location and a predetermined target location of the specified feature in the image; and causing the actuation based on the offset.
10. The system of claim 9, wherein the target location is determined by registering the image to settings of the actuatable optical element during calibration, the calibration comprising: coupling the light from the first optical cores into respective cores of the second optical cores; and coupling back-reflected light from the second optical cores into the first optical cores as the settings of the actuatable optical element are being scanned.
11. The system of claim 1, wherein: the light emitted by the first optical cores is first light; the wavelength of the light emitted by the first optical cores is a first wavelength; the first light source is configured to inject second light having a visible second wavelength different from the first wavelength into a set of source optical cores, the set of source optical cores comprising cores selected from the group consisting of: one or more of the first optical cores and one or more additional optical cores not used to emit the first light; the image is further formed from a reflection of the second light off the end of the optical fiber sensor, such that the image further comprises sub-images of multiple cores of the set of source optical cores; the specified feature in the image comprises sub-images of multiple cores of the optical fiber sensor; and causing the actuation of the actuatable optical element to align the optical paths with the respective second optical cores comprises: causing the actuation to superimpose the sub-images of the multiple core of the optical fiber sensor with the sub-images of the multiple cores of the set of source optical cores.
12. The system of claim 1, wherein: the light emitted by the first optical cores is first light; the wavelength of the light emitted by the first optical cores is a first wavelength; the first light source is configured to inject second light having a visible second wavelength different from the first wavelength into a source optical core selected from the group consisting of: the first optical cores, and one or more additional optical cores not used to emit the first light; the image is further formed from a reflection of the second light off the end of the optical fiber sensor, such that the image further comprises a sub-image of the selected source optical core; the specified feature in the image comprises a sub-image of a core of the optical fiber sensor; and causing the actuation of the actuatable optical element to align the optical paths with the respective second optical cores comprises: causing the actuation to superimpose the sub-image of the core of the optical fiber sensor with the sub-image of the selected source optical core.
13. The system of claim 1, wherein: the reflected or scattered illumination comprises illumination that propagates along the optical paths and is collimated, by the second objective element, into collimated reflected or scattered illumination; the system further comprises: a dichroic mirror disposed in the optical paths between the first objective element and the second objective element, wherein the collimated reflected or scattered illumination is redirected, by the dichroic mirror, away from the optical paths as redirected reflected or scattered illumination; and the focusing element is located in a path of the redirected reflected or scattered illumination and configured to focus the redirected illumination onto the imaging array.
14. The system of claim 13, wherein: the dichroic mirror is disposed between the actuatable optical element and the first objective element; and causing the actuation of the actuatable optical element to align the optical paths with the respective second optical cores comprises: determining an offset between the location of the specified feature in the image and a predetermined target location in the

image; and causing the actuation to reduce the offset.

15. The system of claim 1, wherein: the first light source comprises a source optical fiber, the source optical fiber comprising the first optical cores; the second light source is a first illumination light source, and the illumination wavelength is a first illumination wavelength; the system further comprises: a third light source configured to direct second illumination at an end of the source optical fiber at a second illumination wavelength, such that the second illumination reflects or scatters off the end of the source optical fiber in a direction along the optical paths, to be collimated by the first objective element and then to be redirected by a dichroic mirror away from the optical paths to become redirected second illumination; the second illumination wavelength differs from the first illumination wavelength, and differs from the wavelength of the light emitted by the first optical cores; and the system further comprises: a reflector configured to reflect the redirected second illumination towards the focusing element to cause the reflected redirected second illumination to be focused onto the imaging array, such that an image of the end of the source optical fiber is superimposed with the image of the end of the optical fiber sensor.

16. The system of claim 1, wherein: the system further comprises: an interrogator to analyze reflected light, the reflected light being reflected along a length of the second optical cores of the optical fiber sensor and coupled back into the first optical cores of the first light source; the interrogator is configured to determine a magnitude or amplitude of the reflected light; and the operations further comprise: after an initial alignment of the optical paths with the respective second optical cores based on the location of the specified feature in the image, causing another actuation of the actuatable optical element based on the magnitude or amplitude of the reflected light, to further align the optical paths with the respective second optical cores.

17. The system of claim 1, further comprising: a sensor for sensing, based on a second image of the reflected or scattered illumination, a longitudinal separation between the end of the optical fiber sensor and a focus of the light coupled from the first optical cores into the respective cores of the second optical cores; and a position adjustor configured to adjust a position of the focus to reduce the longitudinal separation.

18. The system of claim 17, wherein: the position adjustor comprises a variable focus lens disposed in the optical paths, and the operations further comprise: adjusting the position of the focus by causing the variable focus lens to adjust a collimation of the light; or the position adjustor comprises an actuator configured to move the second objective element, and the operations further comprise: adjusting the position of the focus by causing the actuator to move the second objective element; or the sensor comprises a bi-prism configured to impart a wedge angle between opposing halves of the reflected or scattered illumination and create duplicate features in the second image, and the operations further comprise: determining, based on a spacing between the duplicate features, the longitudinal separation; or the second light source is configured to illuminate the end of the fiber at a plurality of wavelengths, the sensor comprises a split-field dichroic filter or a chromatically aberrated lens and is configured to create duplicate features at two different wavelengths in the second image, and the operations further comprise: determining, based on a spacing between the duplicate features, the longitudinal separation.

19. A system for directing light emitted by first optical cores of a multicore optical fiber into second optical cores of an optical fiber sensor, the system comprising: a first objective element and a second objective element for directing the light from the first optical cores along respective optical paths; an actuatable optical element disposed between the first objective element and the second objective element; a second light source configured to direct illumination toward an end of the optical fiber sensor, such that the illumination reflects or scatters from the end of the optical fiber sensor, propagates along the optical path, and is collimated by the second object objective element into collimated reflected or scattered illumination, wherein the illumination has an illumination wavelength differing from a wavelength of the light emitted by the first optical cores; a dichroic mirror disposed in the optical paths between the first objective element and the second objective

element and configured to redirect the collimated reflected or scattered illumination away from the optical paths as redirected reflected or scattered illumination; a focusing element and an imaging array located in a path of the redirected reflected or scattered illumination and configured to form, from the redirected reflected or scattered illumination, an image of the end of the optical fiber sensor; and a control system comprising one or more processors, the control system configured to perform operations comprising: determining a location of a specified feature in the image, and causing, based on the location, an actuation of the actuatable optical element to align the optical paths with the respective second optical cores.

20. The system of claim 19, wherein: the dichroic mirror is disposed between the actuatable optical element and the first objective element; and causing the actuation of the actuatable optical element to align the optical paths with the respective second optical cores comprises: determining an offset between the location of the specified feature in the image and a predetermined target location in the image; and causing the actuation to reduce the offset.